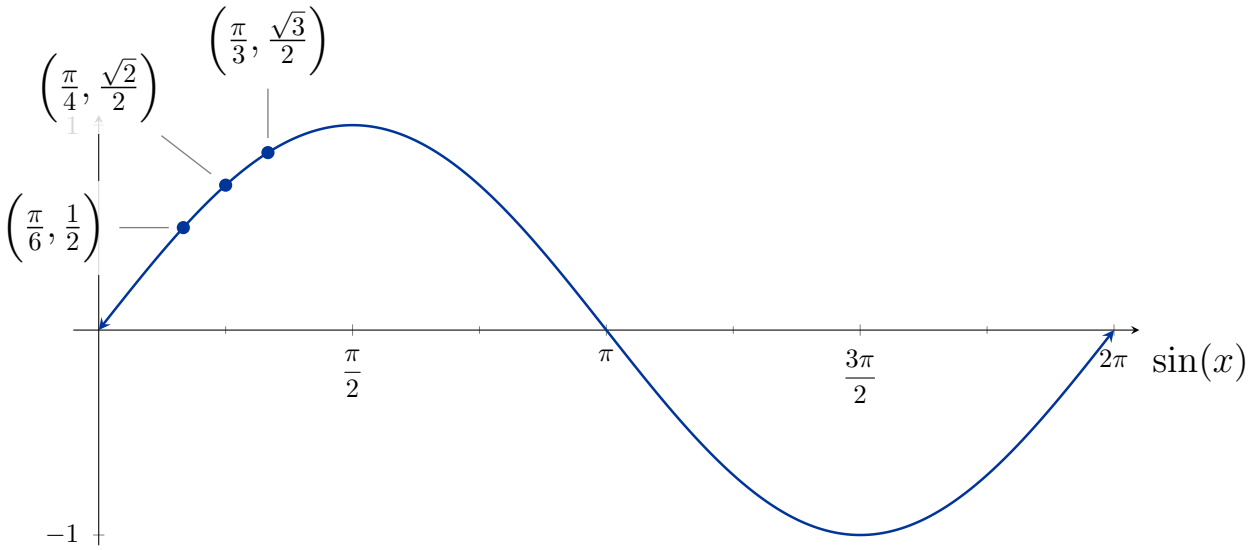
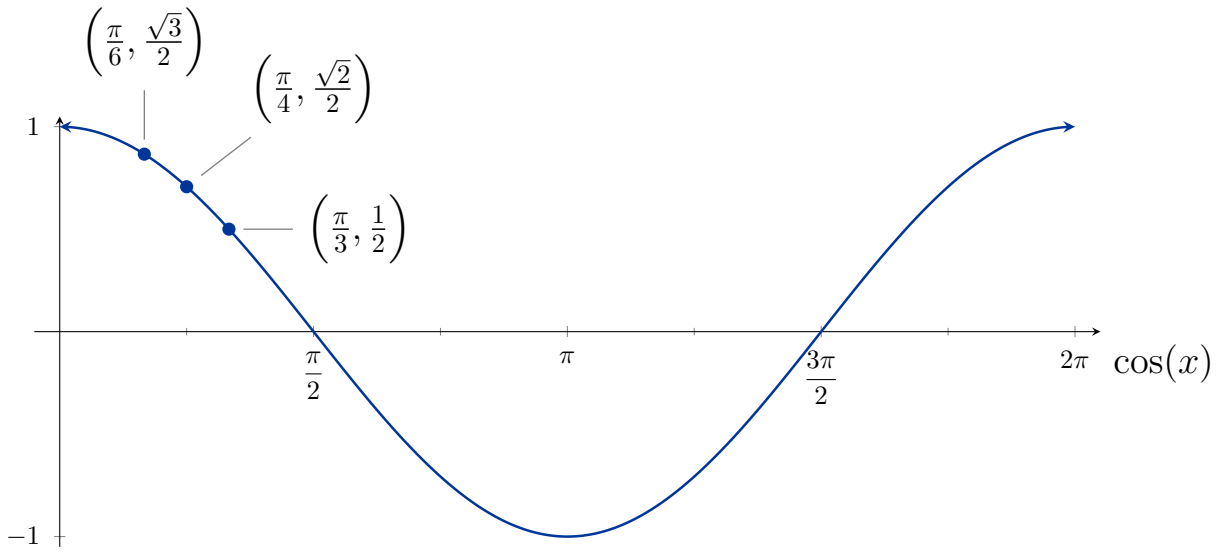
6.1: Graphs of the Sine and Cosine Functions

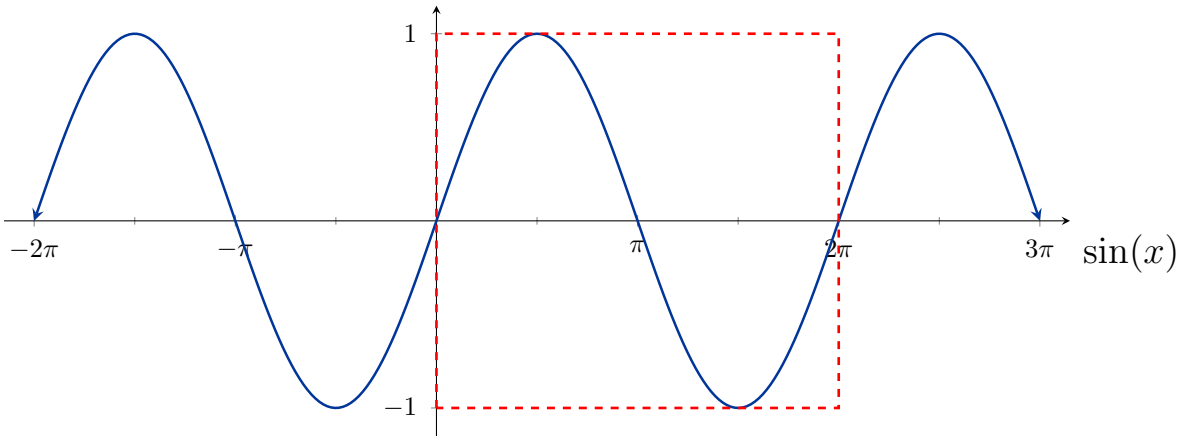


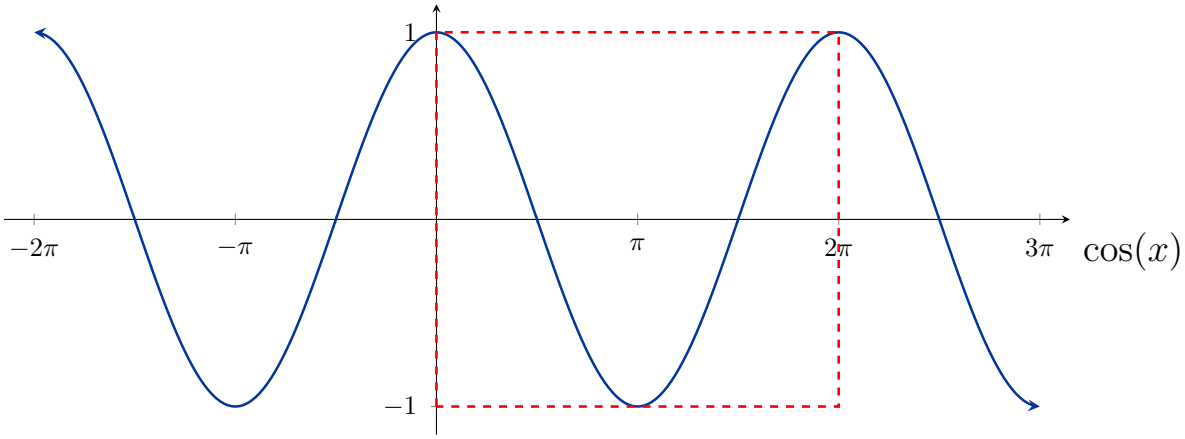


Animated graphs

Recall that

- $\sin(\theta)$ and $\cos(\theta)$ have period 2π,
- $\sin(\theta)$ is an odd function, and
- $\cos(\theta)$ is an even function:





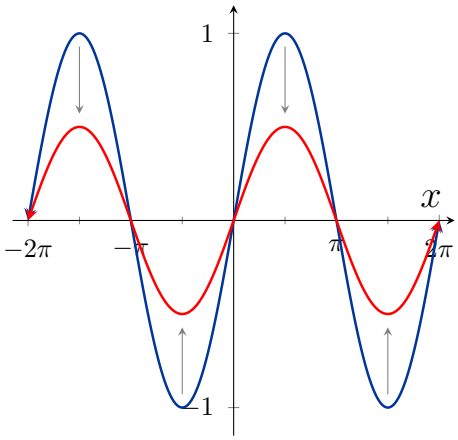
Recall from transformations of functions:

$$af(bx-h)+k$$

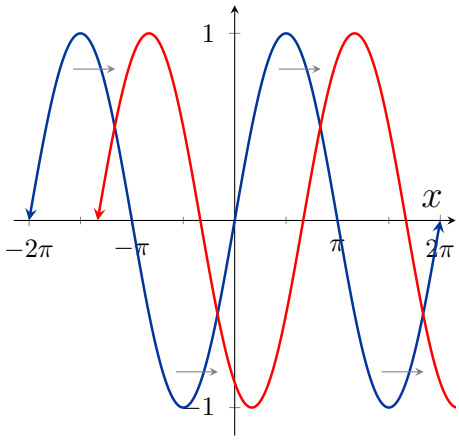
We have that $f(x)$ is

- stretched vertically by a
- stretched horizontally by $1/b$
- shifted right by h
- shifted vertically by k

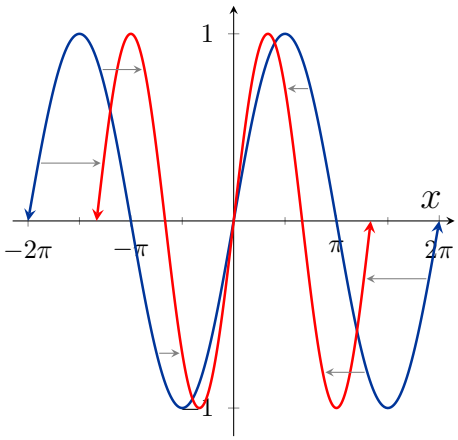
$\sin(x)$ vs. $\frac{1}{2}\sin(x)$



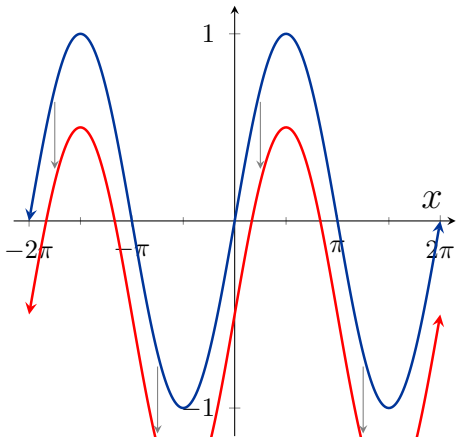
$\sin(x)$ vs. $\sin\left(x-\frac{2\pi}{3}\right)$



$\sin(x)$ vs. $\sin\left(\frac{3x}{2}\right)$



$\sin(x)$ vs. $\sin(x)-\frac{1}{2}$



Definition. (Sinusoidal functions)
A **sinusoidal function** is a function that has the same general shape as a sine or cosine function. The general forms of sinusoidal functions are

$$y = A\sin(Bx - C) + D \quad \text{and} \quad y = A\cos(Bx - C) + D,$$

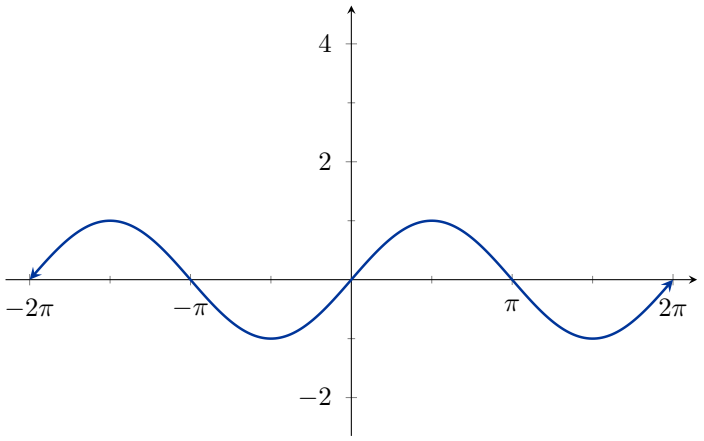
where

- A determines the amplitude,
- B determines the period,
- C determines the phase shift, and
- D determines the vertical shift.

Example. Identify the amplitude, period, phase shift, and vertical shift of

$$3\sin(2x) + 1$$

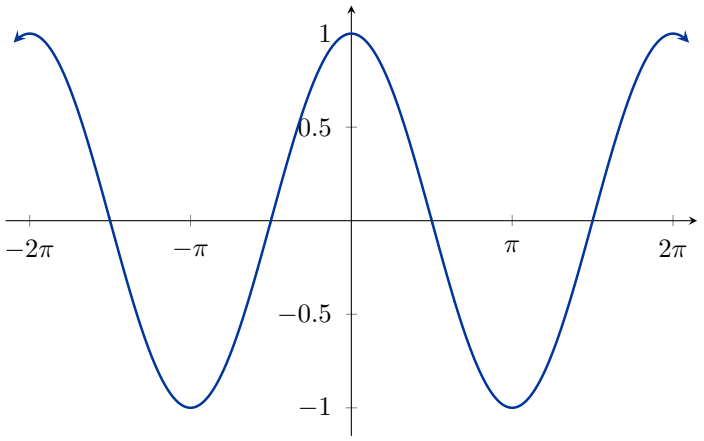
and sketch the graph:



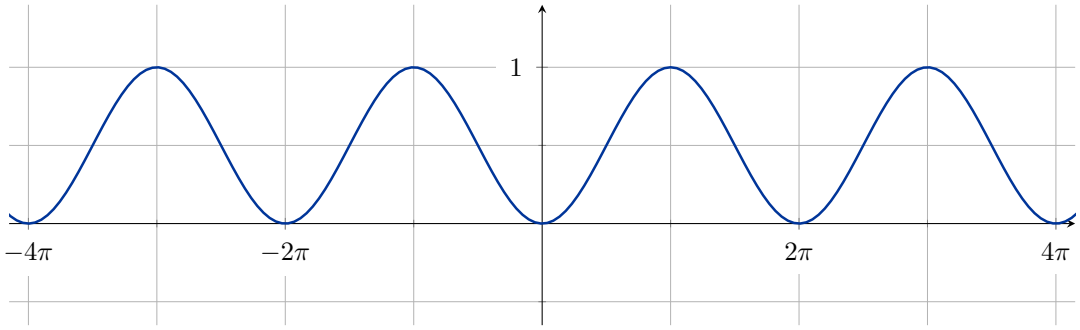
Example. Identify the amplitude, period, phase shift, and vertical shift of

$$\frac{1}{2}\cos\left(\frac{x}{3}-\frac{\pi}{3}\right)$$

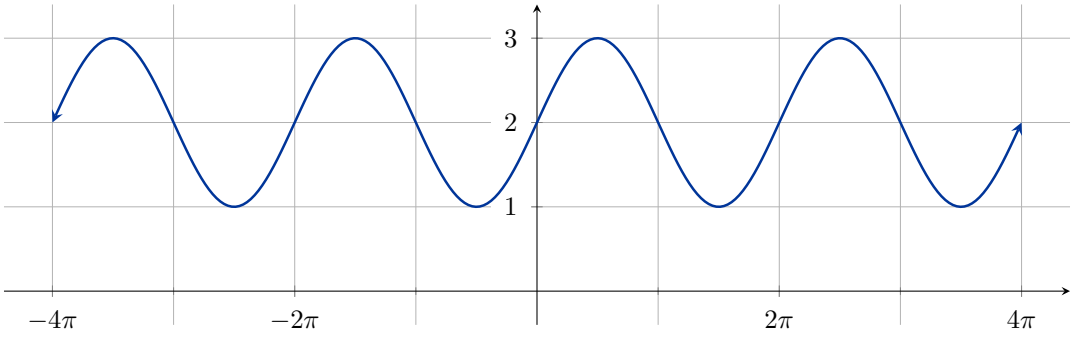
and sketch the graph:



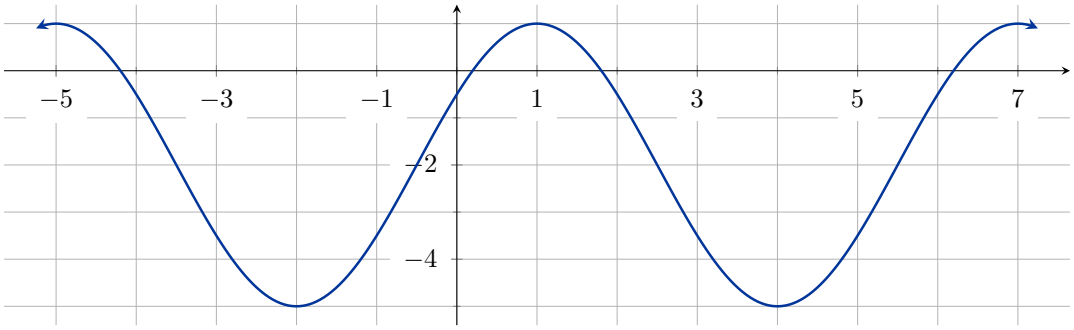
Example. Determine an equation for the following sinusoidal function:



Example. Determine an equation for the following sinusoidal function:



Example. Determine an equation for the following sinusoidal function:



Example. Determine an equation for the following sinusoidal function:

